MTH 1125-Mid-Term Exam-B
Exam time: 90 minutes
Date: April 15, 2026

General instructions:

- No discussion is allowed in the exam room.
- No electronic devices, except calculators, are allowed.

PROBLEM	POINTS	SCORE
A1	5	
A2	5	
A3	5	
A4	5	
A5	5	
A6	5	
A7	5	
A8	5	
B1	10	10
B2	10	10
B3	15	12
B4	15	15
B5	10	10
TOTAL	100	97

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PART A: Multiple choice or Short answer

(5 *mark for each correct answer/choice*)

Problem A1. Find the limit if it exists.

$$\lim_{x \to -\infty} \left(\sqrt{x^2 - 2x + 3} + x\right).$$

Write your answer here: 1

Problem A2. Let $f(x) = e^x + 2\sin x$, and $g(x) = x + 1$ if $x > 0$ and $g(x) = x^2 + 3x + 1$ otherwise. If $h = g \circ f$, then $h'(0) = ?$

Write your answer: $h'(0) =$ 3 .

Problem A3. Evaluate $\lim_{x \to -1} \frac{x^{2026}-1}{x+1}$.

Write your answer here: -2026.

Problem A4. Find $f'(0)$ if it exists:

$$f(x) = \begin{cases} x^4 \sin\frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Circle the best answer.

A.1 B. 0 C. 2 D. It does not exist

Problem A5. Find the limit if it exists

$$\lim_{x \to -1} \frac{|x-2| - |x+3| + x}{x+1}.$$

Circle the best answer.

A. -1 B. $-\infty$ C. 1 D. A,B,C are incorrect

Problem A6. Find all vertical lines of the curve

$$y = \frac{2x^2 - 8x - 10}{(4x-1)(x+1)}$$

Write your answer: $\frac{1}{4}$

Problem A7. If $f'(x) \leq 6$ for all x and $f(2) = -1$ then the largest possible value of $f(8)$ is ..

Circle the best answer.

A. 35 B. 31 C. 50 D. A,B,C are incorrect

Problem A8. Find the 2026th derivative of $f(x) = \sin(3x)$.

Write yours answer here: $-3^{2026} \cdot \sin(3x)$

PART B: WRITE YOUR FULL ANSWERS.

(If the space is not enough, please write on the back and indicate with a phrase like `Problem …continue on…')

Problem B1. Find all values of a and b such that f is differential everywhere:

$$f(x) = \begin{cases} a\cos x + b & \text{if } x < 0 \\ x^2 - 2x - 3 & \text{if } x \geq 0 \end{cases}$$

Problem B2. Find equations of the tangent line at the point $(1,0)$ to the curve

$$y = (\sin x)^{\ln x}$$

(1,1)

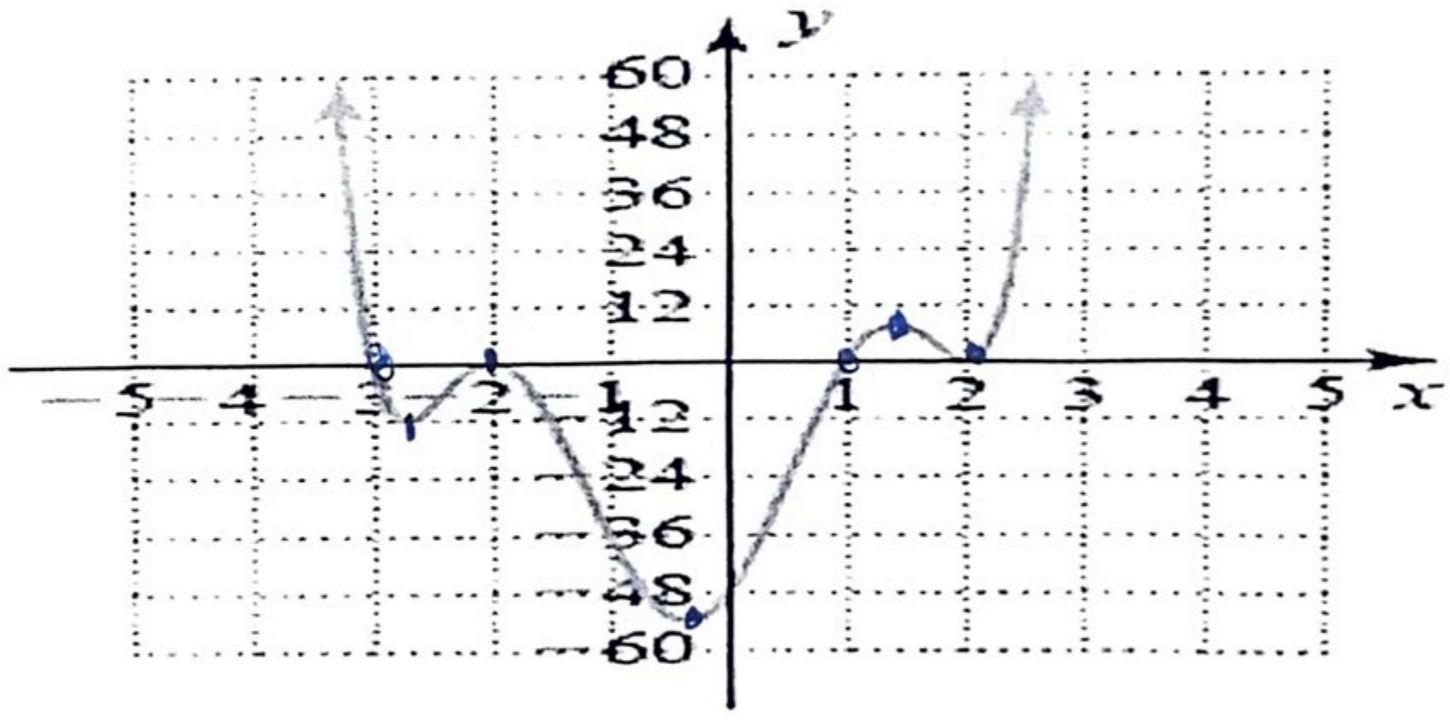
Problem B3. Let $f(x) = x^3 - 3x^2 + mx - 1$, where m is a parameter (a real number).

a. Use the definition of derivative (Section 2.7, Definition 4 - Texbook) to find the derivative of f at the number a.
b. Show that the equation $f(x) = 0$ has at least one real root.
c. Find all m such that $f(x) = 0$ has exactly one real root.

Justify

Problem B4. Given the polynomial $f(x)$ with graph below (Assume all zeroes are real).

a. Sate $\lim_{x \to \infty} f(x)$ and $\lim_{x \to -\infty} f(x)$?
b. How many points where the tangent lines are horizontal? Estimate local minimal/maximal values of f?
c. Use the graph to name the zeroes of f, then state whether their multiplicity is even or odd
d. state the minimum possible degree of f and write one possible function for f in factored form;



Problem B5. Suppose f is a function with the property that $|f(x)| \leq x^4$. Find $f'(0)$?

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Start to write your full solutions (PART B) below.

Problem B1 :

$$f(x) = \begin{cases} a\cos x + b & \text{if } x < 0 \\ x^2 - 2x - 3 & \text{if } x \geq 0 \end{cases}$$

$$\lim_{x \to 0^-} a\cos x + b = \lim_{x \to 0^-} a \cdot \cos 0 + b = a + b.$$

$$\lim_{x \to 0^+} x^2 - 2x - 3 = 0^2 - 2 \cdot 0 - 3 = -3$$

$\Rightarrow$ The function is continuous if:

$$\lim_{x \to 0^-} a\cos x + b = \lim_{x \to 0^+} x^2 - 2x - 3 \Leftrightarrow a + b = -3 \quad (1)$$

$$f'_-(0) = \lim_{h\to 0^-} \frac{f(0+h)-f(0)}{h}$$

$$f'_-(0) = \lim_{h\to 0^-} \frac{a.\cos(h)+b-a.\cos 0-b}{h}.$$

$$f'_-(0) = \lim_{h\to 0^-} \frac{a.\cos(h)-a}{h}$$

$$f'_-(0) = \lim_{h\to 0^-} \frac{a.(\cos(h)-1)}{h} \cdot (a) = 0. \quad (b)$$

$$f'_-(0) = \lim_{x\to 0^-} \frac{f(x)-f(0)}{x} = \lim \frac{a.(\cos x-1)}{x} = 0 \quad (c)$$

$\cos -1 \sim -\frac{x^2}{2} \Rightarrow \frac{\cos x-1}{x} \to 0 \Rightarrow f'_-(0) = 0$. (*).

$$f'_+(0) = \lim_{h\to 0^+} \frac{f(0+h)-f(0)}{h}$$

$$f'_+(0) = \lim_{h\to 0^+} \frac{h^2-2.h-3+3}{h} = \lim_{h\to 0^+} \frac{h^2-2h}{h}.$$

$$= \lim_{h\to 0^+}(h-2) = -2. \quad (a). \quad (**)$$

(*), (**) $\Rightarrow f'_-(0) \neq f'_+(0)$. $(0 \neq -2)$.

$\Rightarrow$ there are no value of a and b. such that f is differential everywhere.

Problem B2:

$$y = (\sin x)^{\ln x}.$$

$$\Rightarrow \ln y = \ln(\sin x)^{\ln x}.$$

$$\Rightarrow \ln y = \ln x . \ln(\sin x).$$

$$\Rightarrow (\ln y)' = (\ln x . \ln(\sin x))'$$

$$\Rightarrow \frac{y'}{y} = \frac{1}{x}\ln(\sin x) + \frac{\cos x}{\sin x}.\ln x.$$

$$\Rightarrow y' = \left(\frac{1}{x}.\ln(\sin x) + \frac{\cos x}{\sin x}.\ln x\right) y.$$

$$\Rightarrow y' = \left(\frac{1}{x}.\ln(\sin x) + \frac{\cos x}{\sin x}.\ln x\right).(\sin x)^{\ln x}.$$

$$\Rightarrow y'(1) = (\sin 1)^{\ln}.\ln(\sin 1).$$

Tangent line function is:

$$y = y'(x - x_0) + y_0.$$

$$y = \ln(\sin 1)(x-1) + 1.$$

Problem B 3:

$$f(x) = x^3 - 3x^2 + mx - 1.$$

a) $$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

$$f'(a) = \lim_{h \to 0} \frac{(a+h)^3 - 3(a+h)^2 + m.(a+h) - 1 - a^3 + 3a^2 - ma + 1}{h}.$$

$$= \lim_{h \to 0} \frac{a^3 - 3a^2h + 3ah^2 + h^3 - 3(a^2 + 2ah + h^2) + ma + mh - 1 - a^3 + 3a^2 - ma + 1}{h}$$

$$= \lim_{h \to 0} \frac{h(3a^2 - 6a + m) + h^2(3a - 3) + h^3}{h}$$

$$= \lim_{h \to 0} 3a^2 - 6a + m + h.(3a - 3) + h^2.$$

$$= 3a^2 - 6a + m.$$

$\Rightarrow f'(x) = 3x^2 - 6x + m.$

b) When $x \to +\infty \Rightarrow f(x) \to +\infty.$

$x \to -\infty \Rightarrow f(x) \to -\infty.$ } -2

$\Rightarrow \exists c$ in order for $f(c) = 0.$

$\Rightarrow f(x) = 0$ has at least one real root.

c) $f'(x) = 0 \Leftrightarrow 3x^2 - 6x + m = 0.$

$\Delta = (-6)^2 - 4.3.m = 36 - 12m.$

In order to find m such that $f(x) = 0$ has exactly one real root,

$\Rightarrow \Delta = 36 - 12m = 0.$

$\Leftrightarrow m = 3.$? -1

Problem B4:

a) Since the graph went up when $x \to \infty$ and $x \to -\infty$.

$\Rightarrow \lim_{x \to \infty} f(x) = \infty$ and $\lim_{x \to -\infty} f(x) = \infty$.

b) There are 5 points where the tangent line are horizontal.

Local minimum / maximum value of f is:

$x \approx -2,8 \rightarrow f = 12$.

$x = -2 \rightarrow f = 0$

$x \approx 0 \rightarrow f = 50$

$x \approx 1,5 \rightarrow f = 10$

$x = 1 \rightarrow f = 0$.

c) When $x = -3 \Rightarrow$ odd.

$x = -2 \rightarrow$ even.

$x = 1 \rightarrow$ odd.

$x \times 2 \rightarrow$ even.

All zeros are $x = -3$; $x = -2$; $x = 1$, $x = 2$.

$\Rightarrow$ The multiplicity is even.

d) The minimum possible degree of f is:

$1 + 2 + 1 + 2 = 6$.

$\Rightarrow$ one possible function for f is:

$f(x) = 3 \cdot (x+3)^1 (x+2)^2 (x-1)(x-2)^2$.

Problem B5:

$$|f(x)| \leq x^4.$$

$$|f(0)| \leq 0 \Rightarrow f(0) = 0.$$

$$f'(0) = \lim_{x \to 0} \frac{f(x) - f(0)}{x} = \lim_{x \to 0} \frac{f(x)}{x} \quad (*)$$

with: $|f(x)| \leq x^4$.

$$\Rightarrow \left| \frac{f(x)}{x} \right| \leq \frac{x^4}{|x|} = x^3.$$

$x^3 \to 0$ as $x \to 0$.

$\Rightarrow$ Using squeeze theorem.

$$\Rightarrow \frac{f(x)}{x} \to 0.$$

$(*) \Rightarrow f'(0) = 0.$

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